# Lakhsma Innerwear Ltd

South Vanganahati, Sreepur, Gazipur
(24.190991, 90.451305)
15th November 2018



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# Observations

# Future extension to be carried out

## Observations: Factory Building

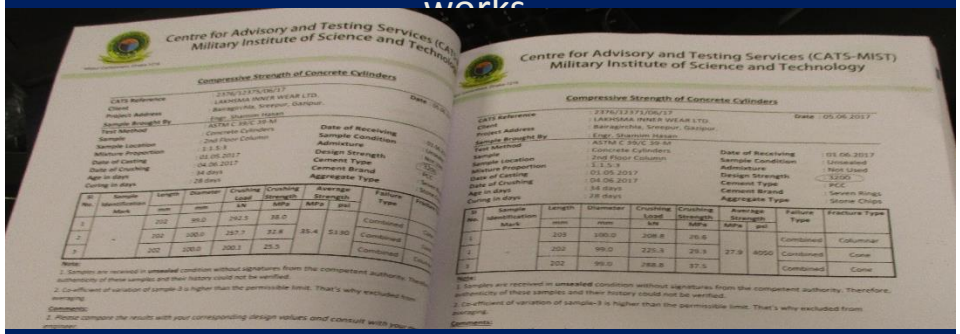


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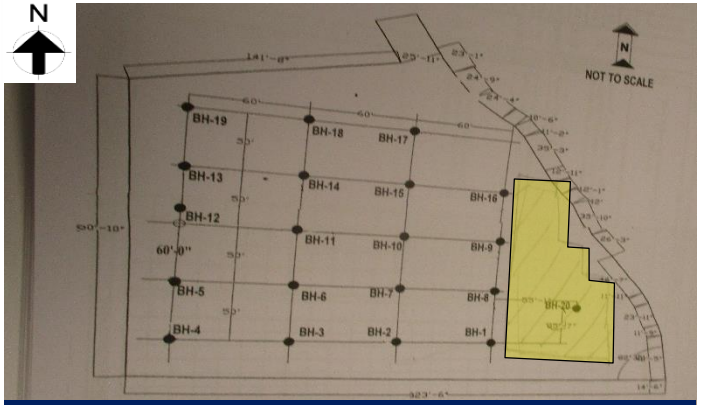


Exposed rebar at roof

Factory has a plan to extend the factory building up to 4-Storey. Also permit drawings available for 4-stroried building. A Detail Engineering Assessment is required before commencement of construction works.




Before Commencement of further construction works evaluate the existing concrete strength either by Cylinder data or 100 mm dia. core.




Bore hole layout

Only one sample BH-20 was considered to calculated the bearing capacity. More samples are required to justify the calculated soil bearing capacity.

**ULTIMATE BEARING CAPACITY (BH 20)**

| Footing | L (m) | B (m) | $D_f$ (m) | $D_f$+B (m) | Avg. SPT-$N_{60}$ upto ($D_f$+B) depth | Effective overburden pressure at ($D_f$+B) depth, $\sigma_0'$ (kN/m$^2$) | Correction Factor, $C_N$ | Avg. SPT-$(N_1)_{60}$ upto ($D_f$+B) depth | $C_u$ (kN/m$^2$) | $N_c$ | $S_c$ | $d_c$ | Water Level (m) | q (kN/m$^2$) | $N_q$ | $s_q$ | $d_q$ | $R_{W1}$ | $q_{ult}$ (ksf) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| F1 | 3 | 3 | 1.95 | 4.95 | 18.89455 | 59.928795 | 1.173014 | 22 | 137.5 | 5.14 | 1.2 | 1.13 | 1.2195 | 30.9 | 1 | 1 | 1 | 0.81 | 20.5 |
| F2 | 2.7 | 2.7 | 1.95 | 4.65 | 19.08 | 57.021795 | 1.189642 | 22 | 137.5 | 5.14 | 1.2 | 1.14 | 1.2195 | 30.9 | 1 | 1 | 1 | 0.81 | 20.8 |
| F3 | 2.25 | 2.25 | 1.95 | 4.2 | 19.37455 | 52.661295 | 1.216245 | 23 | 143.75 | 5.14 | 1.2 | 1.17 | 1.2195 | 30.9 | 1 | 1 | 1 | 0.81 | 22.2 |
| F4 | 1.95 | 1.95 | 1.95 | 3.9 | 19.49455 | 49.754295 | 1.235234 | 24 | 150 | 5.14 | 1.2 | 1.2 | 1.2195 | 30.9 | 1 | 1 | 1 | 0.81 | 23.7 |
| F5 | 2.7 | 2.4 | 1.95 | 4.35 | 19.32 | 54.114795 | 1.20714 | 23 | 143.75 | 5.14 | 1.18 | 1.16 | 1.2195 | 30.9 | 1 | 1 | 1 | 0.81 | 21.6 |
| F6 | 3 | 2.7 | 1.95 | 4.65 | 19.08 | 57.021795 | 1.189642 | 22 | 137.5 | 5.14 | 1.18 | 1.14 | 1.2195 | 30.9 | 1 | 1 | 1 | 0.81 | 20.5 |
| | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | 1.2195 | 30.9 | 1 | 1 | 1 | 0.81 | 22.5 |

Bearing capacity calculation considering Bore Hole-20

# Observations: Factory Building



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# Exposed rebar at roof



Rebar's are left exposed without any corrosion protection. Building engineer is required to provide rust roof coating to the exposed rebar.

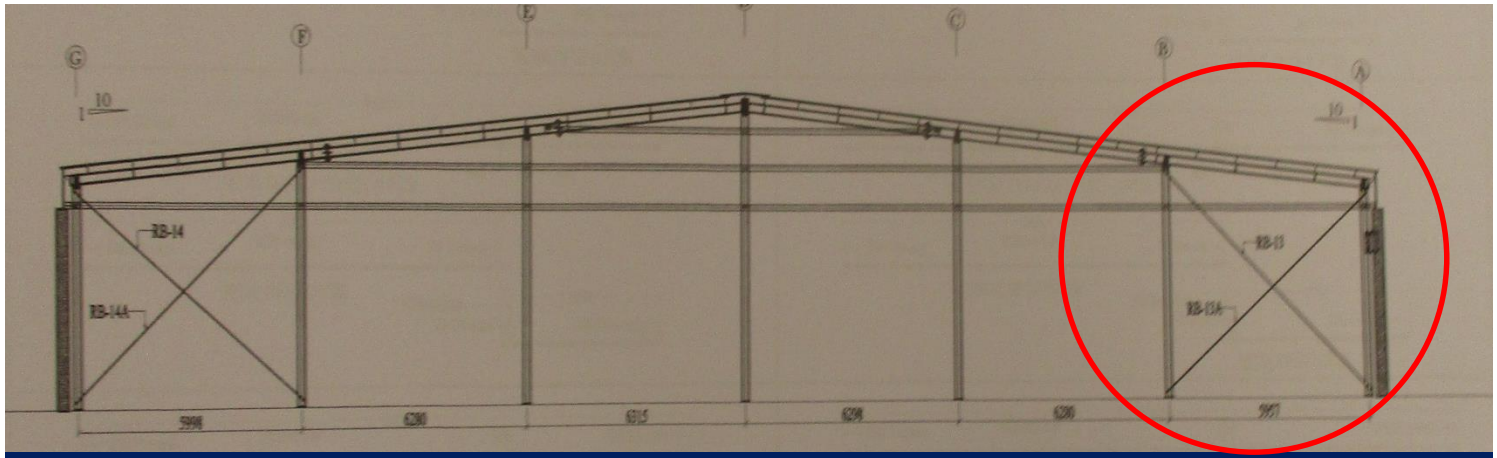
# Observations: Factory Building

# Missing bracing at end frame



## Observations: Sewing Shed

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Wall bracing layout at grid-1




Missing wall bracing

Wall bracing found missing at grid-1. Building engineering is required to install the missing bracing as per specification.



# Observations: Sewing Shed

# Corrosion on steel member



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Corroded steel observed throughout the shed. Building engineer is required to apply rust proof coating.

# Observations: Sewing Shed

# Problems Observed

**Factory Building:**

1: Future extension to be carried out.

2: Exposed rebar at roof.

**Sewing Shed:**

3: Missing bracing at end frame.

4: Corrosion on steel member.



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| Item No. | Observation | Recommended Action Plan | Recommended Timeline |
|---|---|---|---|
| 1 | Future extension to be carried out.(Factory Building) | Additional soil test are required to justify the bearing capacity of soil for the existing structure. | **6-weeks** |
| 2 | Future extension to be carried out.(Factory Building) | Based on the additional geotechnical investigation report review the foundation adequacy. | **6-weeks** |
| 3 | Future extension to be carried out.(Factory Building) | For vertical extension, complete Detail Engineering Assessment (DEA) addressing the observations. | **6-months** |
| 4 | Future extension to be carried out.(Factory Building) | Vertical extension to be carried out as per recommendation of DEA. | **6-months** |
| 5 | Exposed rebar at roof | Apply rust proof coating to the exposed rebar. | **6-weeks** |
| 6 | Missing bracing at end frame. (Sewing Shed) | Installed missing bracing as per specification. | **6-weeks** |
| 7 | Corrosion on steel member. (Sewing Shed) | Apply rust proof coating on steel member. | **6-weeks** |